On the learning representation of LLM : A concept-oriented study

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17$^{\text{th}}$ March, 2025

MATHÉMATIQUES VISION APPRENTISSAGE

école normale supérieure paris–saclay

1. Introduction
2. Observing Representation
3. Going Into Representation
4. Conclusion

Classic Machine Learning

$$\min_{\theta\in\Theta} \mathcal{C}(\theta) := \mathbb{E}_{(x,y)\sim\mathcal{D}}\left[\mathrm{C}(f(\theta;x),y)\right] \tag{ML}$$

- θ: model parameters, Θ: parameter space
- $f(\theta;x)$: model function, $\mathcal{C}$: cost function
- $\mathcal{D}$: data distribution, (x,y): input-target pairs

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SGD Discrete Update:

$$\theta_{t+1} = \theta_t - \eta \cdot \mathbb{E}_{(x,y)\sim\mathcal{B}}(\nabla_\theta \mathrm{C}(f(\theta_t; x), y)) \tag{1}$$

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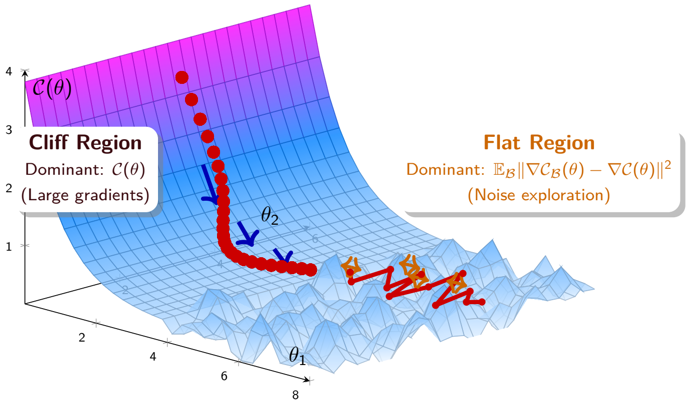
SGD Implicit Regularization [1]:

$$\mathcal{C}_{\mathrm{SGD}}(\theta) = \mathcal{C}(\theta) + \frac{\eta}{4}\|\nabla\mathcal{C}(\theta)\|^2 + \frac{\eta}{4}\mathbb{E}_{\mathcal{B}}\|\nabla\mathcal{C}_{\mathcal{B}}(\theta) - \nabla\mathcal{C}(\theta)\|^2 \quad (2)$$

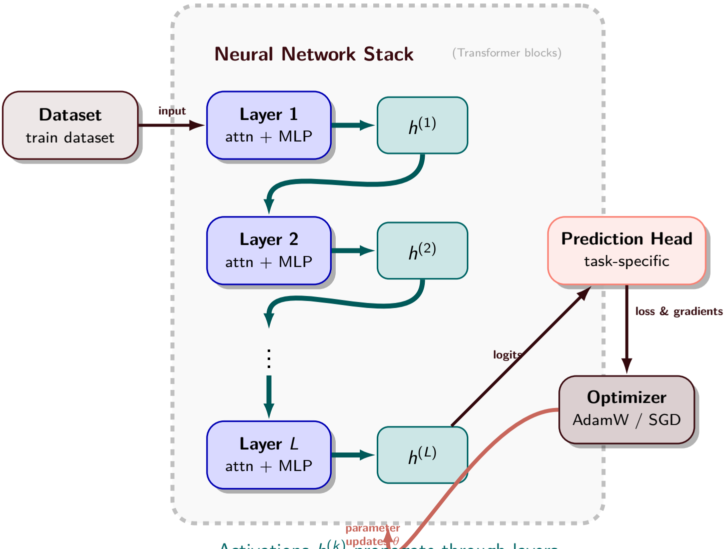
Where $\mathcal{C}_{\mathcal{B}}(\theta)$ is the cost on the mini-batch $\mathcal{B}$.

SGD Dynamics: From Cliff to Flat Landscape

SGD Trajectory: *Smooth → Erratic*

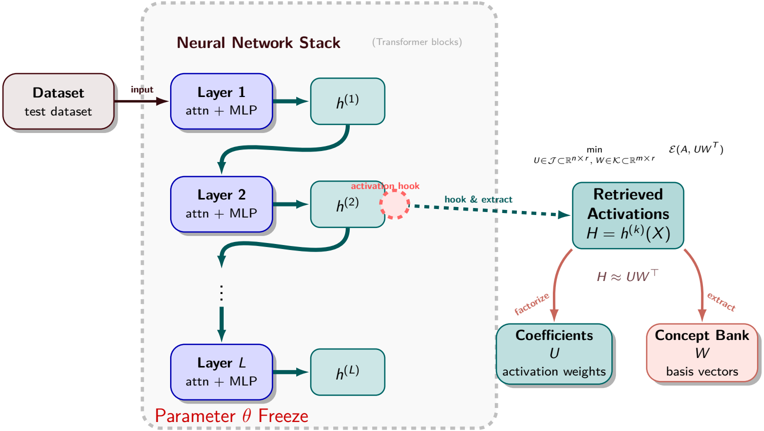


Deep learning - pipeline



Activations $h^{(k)}$ propagate through layers

Concept-Oriented Mechanistic Interpretability



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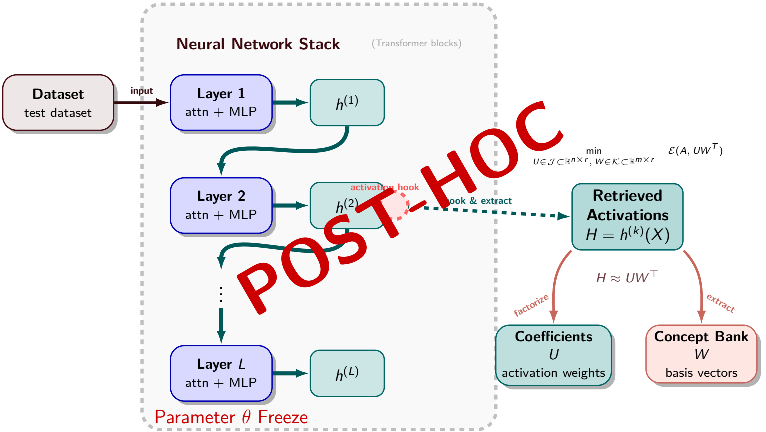
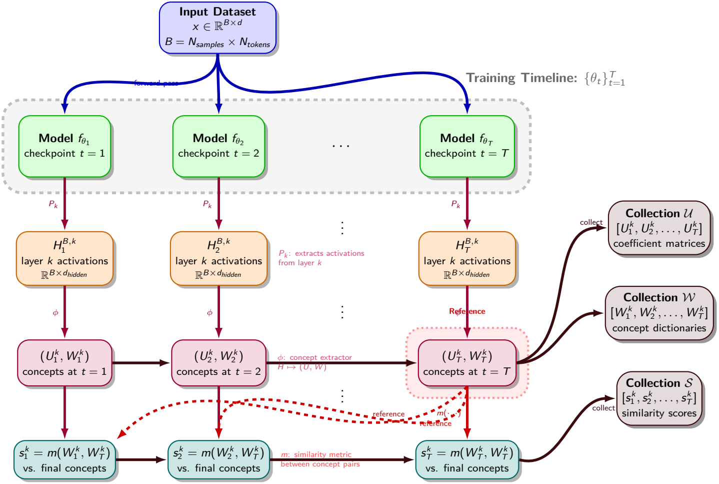


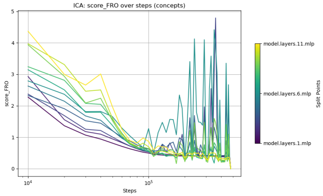
Algorithm: Concept Evolution Across Training Checkpoints

Algorithm: Compute concept representations across checkpoints and similarity to final-step concepts.

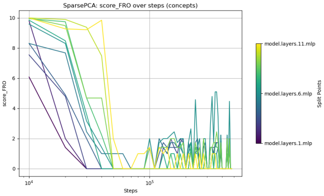


Inputs: x (dataset), f_θ (model), $\{\theta_t\}_{t=1}^{T}$ (checkpoints), $\{P_k\}$ (projectors), ϕ (extractor), m (metric)

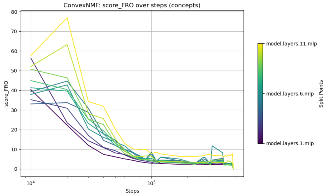
Empirical Learning Dynamics



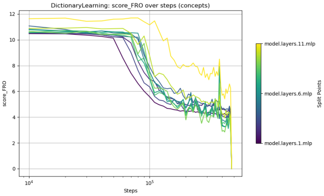
ICA Concepts



Sparse PCA Concepts

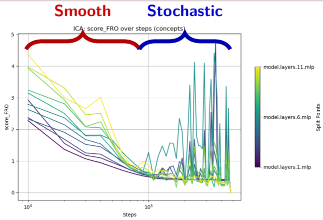


Convex NMF Concepts



Dictionary Learning Concepts

Empirical Learning Dynamics



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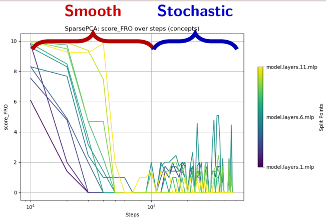

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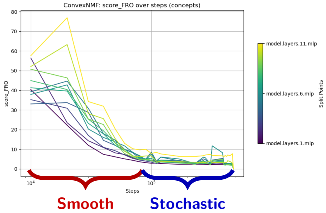

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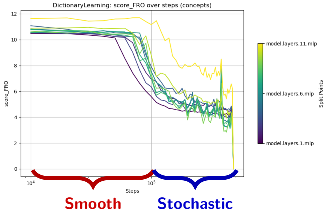
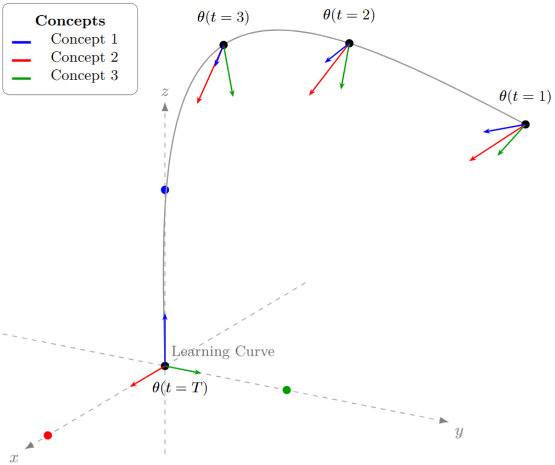
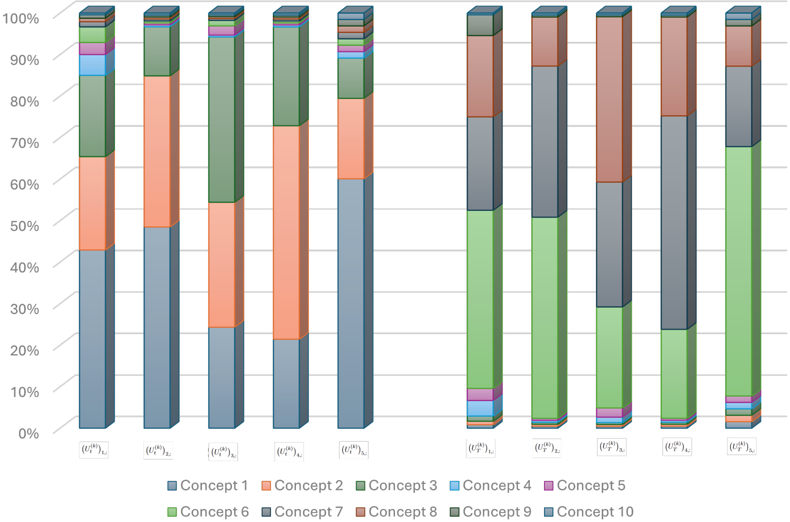


Dictionary Learning Concepts

The Impossible Visualization of High-Dimensional Concepts



The Concept Distribution Evolution: An Example Histogram



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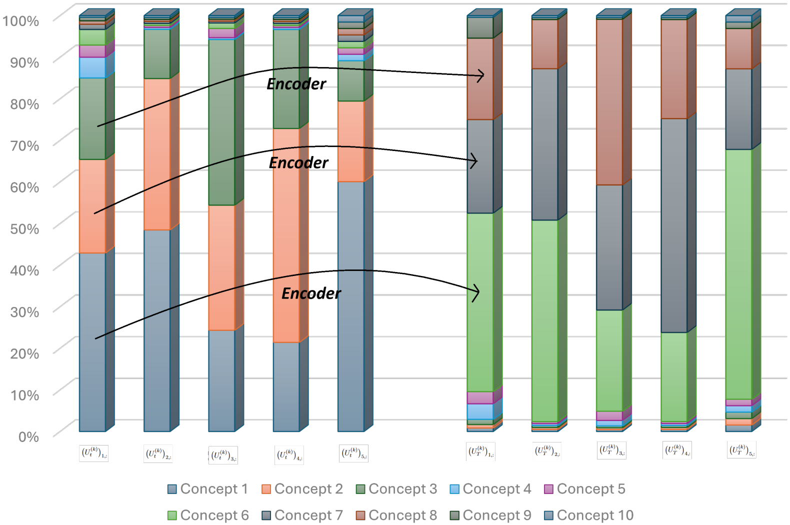
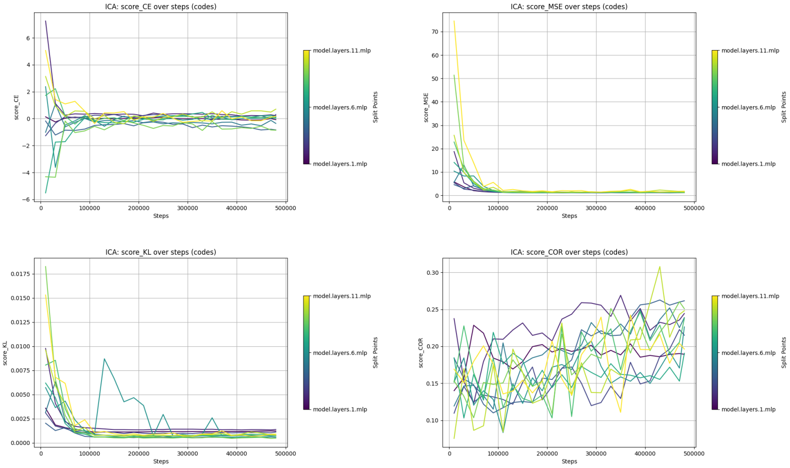
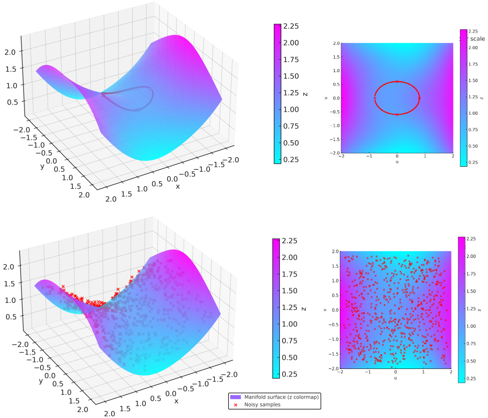


The Concept Distribution Evolution: Learning Dynamics

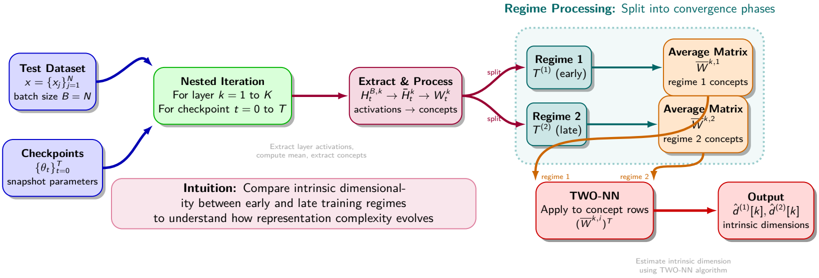


What is a Manifold ?



Intuition: A manifold is a surface that *locally* looks like Euclidean space. Even though the surface is curved in 3D, any path on it can be described using only **2 parameters** -> 2 degrees of freedom.

The Concepts Manifold: Estimating Intrinsic Dimension



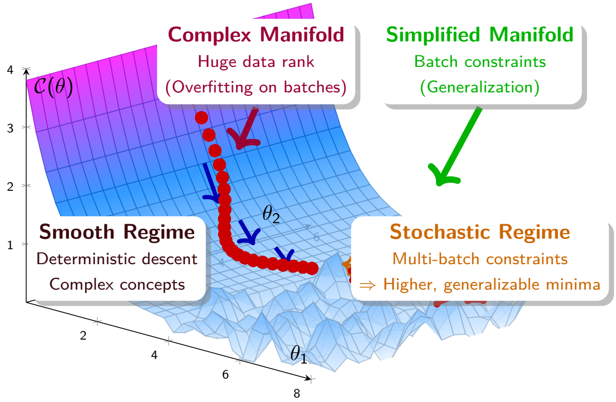
Intuition: Compare intrinsic dimensionality between early and late training regimes to understand how representation complexity evolves

The Concepts Manifold: Results Overview

Layer	Non-sparse methods						Sparse methods			
	SVD		ICA		Semi-NMF		Dict. Learning		K-Means	
	First	Second	First	Second	First	Second	First	Second	First	Second
1	34.91	24.70	46.98	31.09	34.06	24.95	13.17	15.25	11.10	15.57
2	28.72	25.05	37.94	23.79	27.04	27.97	12.35	13.58	9.61	9.98
3	35.84	34.02	23.14	8.55	22.77	11.29	9.76	10.19	3.05	2.52
4	40.35	34.27	17.54	16.51	20.01	12.73	11.20	8.03	3.54	4.01
5	42.18	37.83	22.85	20.86	21.58	18.38	8.67	7.83	3.39	3.91
6	46.52	36.01	14.58	9.99	10.42	8.76	7.28	7.79	3.18	3.13
7	36.96	37.08	29.10	21.27	15.24	24.22	7.33	6.00	6.32	6.30
8	49.08	38.95	20.17	21.99	40.13	22.32	9.04	6.04	8.58	8.67
9	48.40	55.21	20.69	14.67	20.96	13.78	10.69	8.21	8.45	17.47
10	75.07	57.50	44.47	37.67	34.07	32.38	10.64	7.12	7.74	14.02
11	68.98	55.10	20.36	20.79	24.42	23.50	8.30	5.69	4.78	6.70
Mean	46.09	39.61	27.07	20.65	24.61	20.03	9.86	8.70	6.34	8.39

Estimated intrinsic dimension per layer and regime. For each method we report the block-mean intrinsic dimension in the **First** and **Second** regimes. The **gray** cell indicates which regime (first or second) yields the *lower* intrinsic dimension for that layer-method pair. The final row (**Mean**) gives the average intrinsic dimension across layers.

The Concepts Manifold: Interpretation



Satisfying diverse batch gradients forces higher minima that generalize better, leading to manifold dimension reduction through concept simplification

Conclusion

Why larger models generalize better when trained longer ?

Maybe because:

Stochastic regime acts as concept refinement process
$\Rightarrow$ Manifold dimension reduction $\Rightarrow$ Better generalization

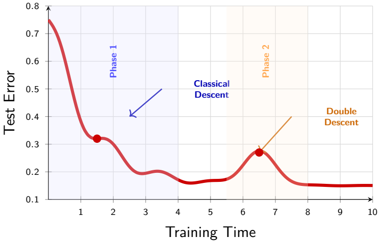
Future Directions:

- Decomposer-aware intrinsic dimension estimators for structured representations
- Deeper LLM analysis across model sizes, other architectures, and tasks
- Novel concept-based convergence approaches
- Extension to multimodal and continual learning settings

5 Annex

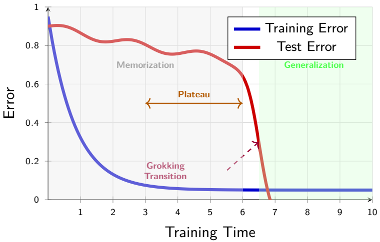
The Big Question: Why does deep learning generalize so well?

Double Descent Phenomenon [2]



Test error initially decreases, then increases, then decreases again

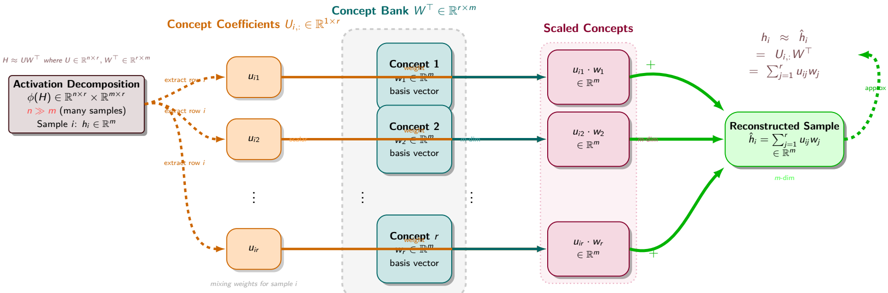
Grokking Phenomenon [3]



Test error plateaus during memorization, then suddenly drops during generalization

Intuition: Both phenomena challenge classical learning theory. **Double descent** shows that larger models can generalize better even when appearing to overfit, while **grokking** demonstrates that generalization can emerge suddenly after prolonged memorization phases.

Concept-Based Sample Reconstruction



Intuition: Linear combination of r concept vectors weighted by sample-specific co-efficients reconstructs activation

Unsupervised Concept Extractor Methods

General Optimization Framework:

$$\phi(h_\theta(x)) \in \arg\min_{W\in\mathcal{K}} \left(\min_{U\in\mathcal{J}} \mathcal{E}\left(h_\theta(x), UW^\top\right)\right)$$

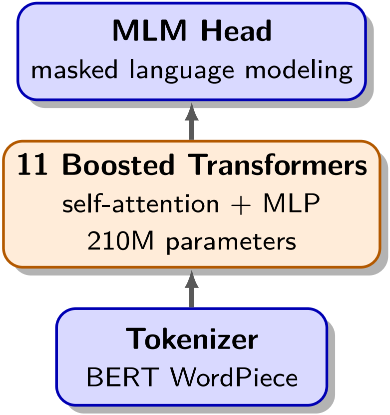
Method	$\mathcal{E}(A,B)$	$\mathcal{J}$ **(Encoded Activations)**	$\mathcal{K}$ **(Concept Dictionary)**
		NMF Family	
NMF (vanilla)	$\|A-B\|_F^2$	$\mathbb{R}_+^{n\times r}$	$\mathbb{R}_+^{m\times r}$
Convex NMF	$\|A-B\|_F^2$	$\{U = AH,\ H \geq 0,\ \mathbf{1}_m^\top H = \mathbf{1}_r^\top\}$	$\mathbb{R}_+^{m\times r}$
Semi-NMF	$\|A-B\|_F^2$	$\mathbb{R}^{n\times r}$	$\mathbb{R}_+^{m\times r}$
		Statistic / SVD–ICA Family	
SVD	$\|A-B\|_F^2$	$\{U^\top U = \Sigma(A)\}$	$\{V^\top V = \Sigma(A)\}$
PCA	$\|A-B\|_F^2$	$\{U : \mathbf{1}_n^\top U = 0\}$	$\{W^\top W = I_r\}$
Sparse PCA	$\|A-B\|_F^2$	$\{U : \mathbf{1}_n^\top U = 0,\ \|U\|_1 \leq \tau\}$	$\{W^\top W = I_r\}$
ICA	$\|A-B\|_F^2$	$\mathcal{U}_{\mathrm{ICA}}(A, \boldsymbol{\tau})$	$\mathbb{R}^{m\times r}$
		Sparse / Dictionary Family	
Dictionary Learning	$\|A-B\|_F^2$	$\{U : \|U\|_1 \leq \tau\}$	$\mathbb{R}^{m\times r}$
Sparse Auto-Encoder	$\|A-B\|_2^2$	$\{U = E_\theta(A)\}$	$\{W : \|W\|_1 \leq \tau\}$
k-Means	$\|A-B\|_F^2$	$\{U \in \{0,1\}^{n\times r},\ U\mathbf{1}_r = \mathbf{1}_n\}$	$\mathbb{R}^{m\times r}$

5 Annex
- Extending Introduction
- Extending Observing Representation
- Parameter Convergence Dependance
- Extending Into Representation
- Contribution

6 References

Empirical Learning Dynamics: Setup

Large Language Model:
EuroBERT-210m[4]



Example Datasets:

RottenTomatoes

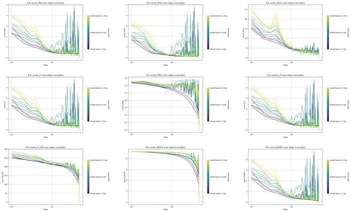
Movie and TV series reviews with critic ratings and sentiment labels.

Wikipedia

Encyclopedia articles covering diverse topics and knowledge domains.

Setup: Analyze concept dynamics in EuroBERT during training

Empirical Learning Dynamics



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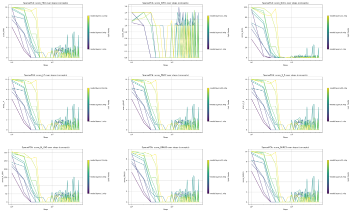

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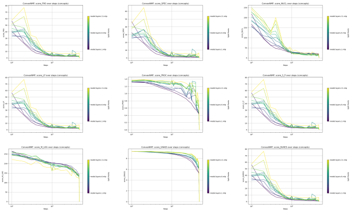

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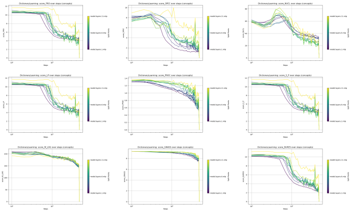

Dictionary Learning Concepts

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Extending Introduction

Extending Observing Representation

Parameter Convergence Dependance

Extending Into Representation

Contribution

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Learning Dynamics : Parameter Convergence

- **Assumption A:** The input set $\mathcal{X}$ is bounded and finite-dimensional.
- **Assumption B:** The parameter set Θ is compact and finite-dimensional.
- **Assumption C:** The model f_θ is a Transformer-based LLM with activation functions $\{\sigma_i\}_{i=1}^{N}$ that are locally Lipschitz.

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Extractor Algorithm-Based Estimator: $\hat{\phi} : A \mapsto (\hat{U}(A), \hat{W}(A))$

(i) The algorithm uses only a fixed *maximum iteration* stopping criterion.

(ii) Each computational step of the algorithm is locally Lipschitz with respect to its inputs.

(iii) All stochastic initializations are determined prior to execution.

Learning Dynamics : Parameter Convergence

Theorem: Under Assumptions A, B, C, and the Extractor Algorithm-Based Estimator conditions (i), (ii), (iii), we have that for any layer k and input $x \in \mathcal{X}$, for all training steps t :

$$d(\hat{\phi}(P_k(f_{\theta_t}, x)),\ \hat{\phi}(P_k(f_{\theta_T}, x))) \leq L_d L_\phi L_f(x) \|\theta_t - \theta_T\|.$$

Which implies,

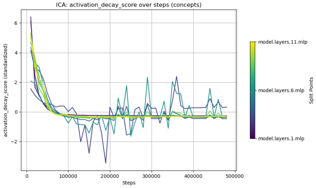
$$s_t^k = O(\|\theta_t - \theta_T\|).$$

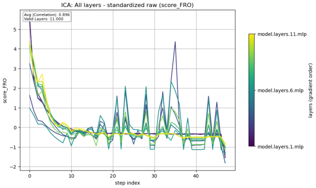
This means that the **learning dynamics** of the **concept** representations are directly **controlled** by the **parameter convergence** of the model during training.

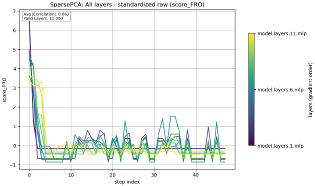
Learning Dynamics : Signal Analysis

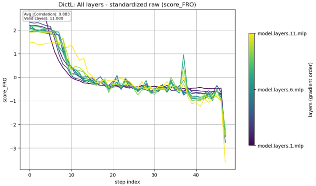
Activation decay signal for each layer k:

$$a_t^k = \|P_k(f_{\theta_t}, x) - P_k(f_{\theta_T}, x)\|_F = \|H_t^{(k)} - H_T^{(k)}\|_F,$$









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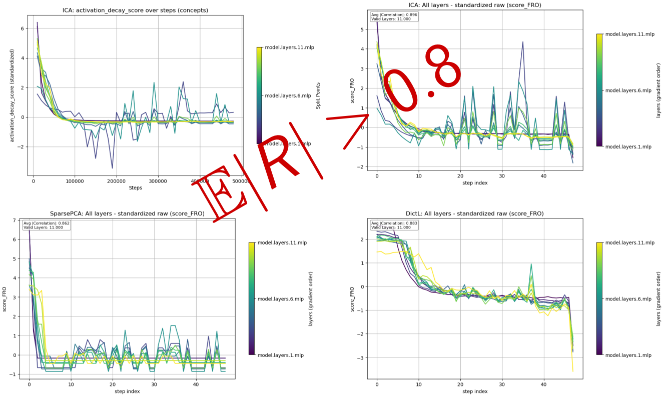


Learning Dynamics : Parameter Convergence - 1

Lipschitz Continuity - A function $f : \Omega \to \Gamma$ is Lipschitz continuous if and only if there exists $L > 0$ such that, for all $x, y \in \Omega$,

$$\|f(y) - f(x)\|_\Gamma \leq L\|y - x\|_\Omega.$$

Here, $\|\cdot\|_\Omega$ and $\|\cdot\|_\Gamma$ denote norms on the spaces Ω and Γ respectively.

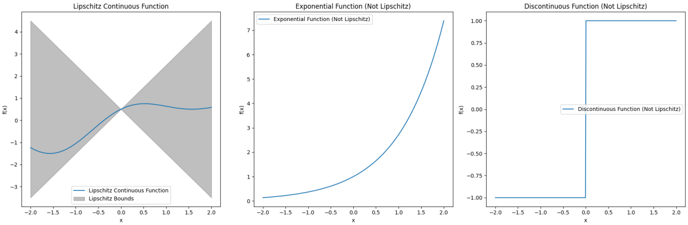
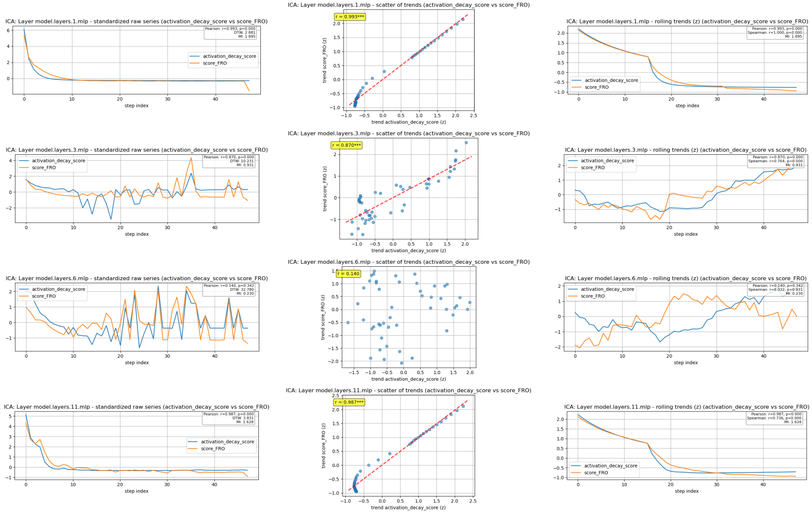
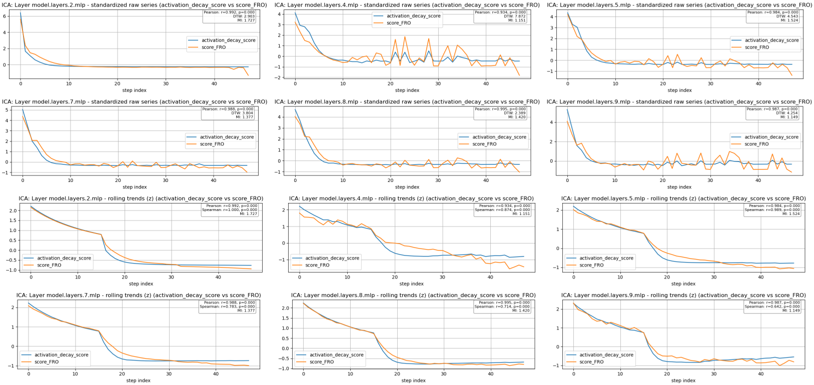


Figure: Lipschitz Visualization (GitHub:Raphael-Bernas).

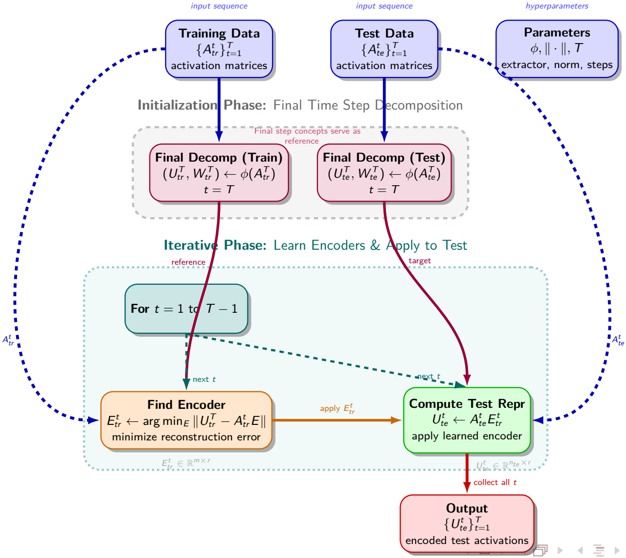
Learning Dynamics : Signal Analysis - 2



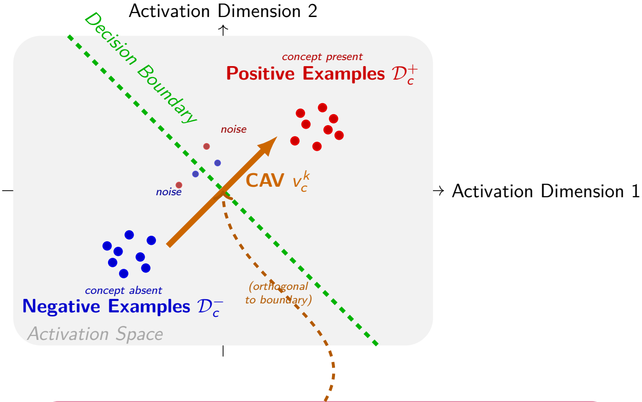
Learning Dynamics : Signal Analysis - 2



Algorithm: Activation-based Representation Learning Algorithm



Concept Activation Vectors (CAV): The Method [5]



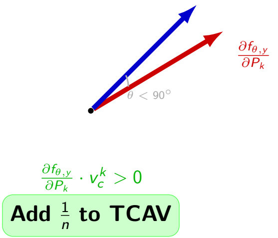
CAV Learning: Find vector v_c^k that best separates $\mathcal{D}_c^+$ and $\mathcal{D}_c^-$ in the activation space using supervised learning (e.g., logistic regression)

Testing with CAV: The Method [5]

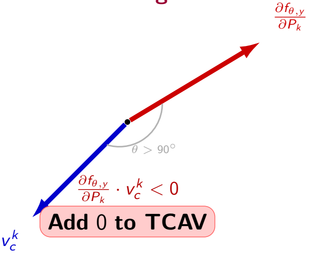
$$\mathrm{TCAV}^k_{y,c} = \mathbb{E}_{x\sim\mathcal{D}_y}\left[\mathbb{I}\left(\frac{\partial f_{\theta,y}(x)}{\partial P_k(f_\theta, x)} \cdot v_c^k > 0\right)\right],$$

where $\mathbb{I}(\cdot)$ denotes the indicator function. It measures the fraction of inputs for which moving in the direction of the concept increases the probability of predicting class y.

Case 1: Aligned



Case 2: Misaligned

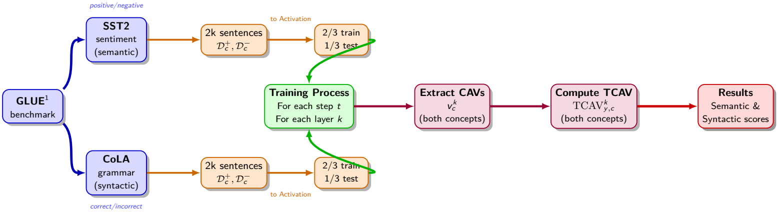


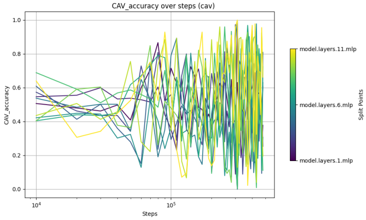
TCAV: Regime-wise analysis

	Syntactic			Semantic		
Layer	**First**	**Second**	**Overall**	**First**	**Second**	**Overall**
1	**0.597 (0.145)**	0.440 (0.115)	0.514 (0.124)	0.571 (0.137)	**0.581 (0.109)**	0.561 (0.142)
2	0.514 (0.077)	**0.594 (0.184)**	0.498 (0.184)	0.535 (0.186)	**0.635 (0.179)**	0.536 (0.195)
3	**0.543 (0.146)**	0.480 (0.150)	0.473 (0.202)	0.329 (0.128)	**0.339 (0.191)**	0.455 (0.201)
4	**0.385 (0.180)**	0.341 (0.237)	0.445 (0.220)	0.423 (0.183)	**0.577 (0.294)**	0.514 (0.220)
5	0.483 (0.176)	**0.506 (0.264)**	0.496 (0.224)	**0.441 (0.235)**	0.367 (0.201)	0.441 (0.205)
6	0.342 (0.116)	**0.574 (0.250)**	0.443 (0.221)	0.308 (0.126)	**0.528 (0.270)**	0.427 (0.219)
7	**0.606 (0.156)**	0.484 (0.221)	0.479 (0.240)	**0.650 (0.205)**	0.277 (0.232)	0.463 (0.252)
8	**0.541 (0.087)**	0.337 (0.203)	0.504 (0.228)	0.502 (0.175)	**0.541 (0.248)**	0.473 (0.232)
9	0.501 (0.168)	**0.725 (0.162)**	0.514 (0.251)	**0.663 (0.173)**	0.572 (0.291)	0.541 (0.241)
10	0.512 (0.230)	**0.619 (0.247)**	0.566 (0.258)	**0.684 (0.162)**	0.491 (0.284)	0.548 (0.263)
11	0.593 (0.172)	**0.657 (0.275)**	0.573 (0.284)	0.606 (0.277)	**0.678 (0.190)**	0.672 (0.241)
$\boldsymbol{\Sigma\Delta}$		**+0.142**			**-0.127**	
$\boldsymbol{\Sigma\omega\Delta}$		**+0.125**			**-0.292**	

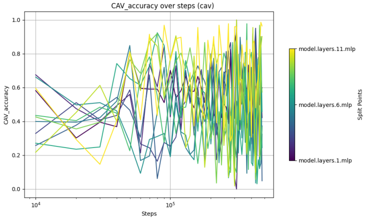
- **Shaded**: Higher regime mean for each layer/modality
- **Blue**/**Red**: Lower/Higher overall means between overall columns
- $\Sigma\Delta$: Simple sum of (Second - First) regime differences
- $\Sigma\omega\Delta$: Variance-weighted aggregate of regime changes

TCAV: Learning Dynamics





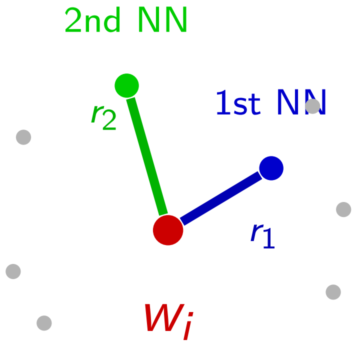
Syntactic



Semantic

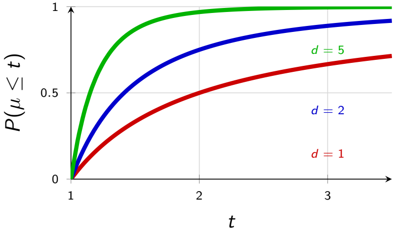
Intrinsic Dimension Estimation: The TWO-NN Theory - 1 [6]

TWO-NN Concept



$$\mu_i = \frac{r_2(w_i)}{r_1(w_i)}$$

Theory: CDF Relationship

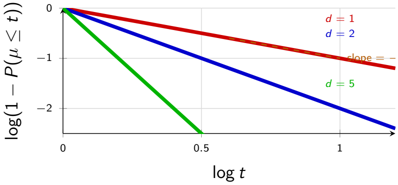


$$P(\mu \leq t) = 1 - t^{-d}$$

Core Intuition: The ratio $\mu = r_2/r_1$ follows a theoretical distribution that depends on intrinsic dimension d

Intrinsic Dimension Estimation: The TWO-NN Theory - 2 [6]

Estimation Method: Linear fitting in log-log space



Algorithm Steps:

1. For each point w_i, compute distances to 1st and 2nd nearest neighbors
2. Calculate ratio $\mu_i = r_2(w_i)/r_1(w_i)$ for all points
3. Sort ratios and compute empirical CDF: $P(\mu \leq t)$
4. Transform to log-log scale: $\log(1 - P(\mu \leq t))$ vs $\log t$
5. Fit linear regression and extract slope $= -\hat{d}$

EuroBERT Training Loss Curve

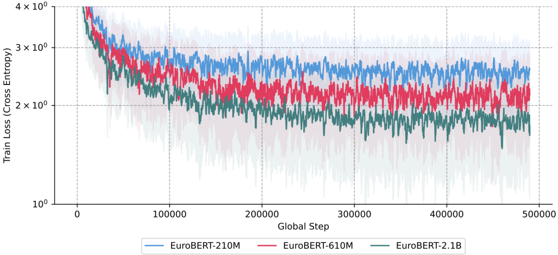


Figure: EuroBERT training loss curve for all family models [4]

5 Annex

Contribution

Main Contributions:

- **Unified MI Framework:** Proposed general mechanistic interpretability framework unifying classical methods
- **Two-Regime Confirmation:** Empirically confirmed dual training phases in LLMs beyond noisy loss curves
- **Concept Distribution Stability:** Showed concept distributions remain stable despite individual vector noise
- **Semantic Validation:** TCAV analysis confirmed regimes exist at human-interpretable concept level even if need more refined concepts
- **Manifold Simplification:** **Key finding:** Intrinsic dimension decreases from regime 1 $\to$ 2

5 Annex

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